

ATC User Guide

Advanced Trigonometry Calculator User Guide

This is a basic repository-level user guide for Advanced Trigonometry Calculator (ATC). It does not replace the full ATC user guide distributed with the application.

For current technical behavior, see:

- docs/en/User_Guide_Full.md
- docs/pt-PT/User_Guide_Full.md
- docs/ATC_2.1.7_DOCUMENTATION.md
- docs/RELEASE_2.1.7.md
- tests/ATC_USER_GUIDE_COVERAGE.md

What ATC Is

ATC is a free, open-source Windows command-line mathematical application. Users enter expressions or documented ATC commands, and ATC prints results in the console.

ATC is focused on practical numerical workflows and documented command behavior.

Quick Start

```
2+2
sin(pi/2)
mode
solve equation(x^2-5*x+6)
solver(x+2)
create matrix(foo\2\2\3)
see variables
exit
```

Remember:

- `_` is ATC's documented negative marker in many command forms and outputs;
- `#0`, `#1` and later indexes refer to previous results;
- direct expressions calculate immediately;
- commands start named workflows;
- guided modules open interactive menus.

What ATC is not

ATC deliberately focuses on practical command-driven mathematical computation instead of trying to be a universal computer algebra system.

- ATC is not a full general-purpose CAS like Mathematica, Maple, SageMath, or SymPy.

- ATC is not designed to symbolically prove arbitrary mathematical theorems.

- ATC is not intended to replace domain-specific professional engineering validation tools.

- ATC focuses on numeric computation, command-driven workflows, educational use, implemented symbolic/numeric hybrid features, and fast local execution.

- Some symbolic behavior exists, but only where explicitly supported and

tested.

Basic Expressions

```
2+3*4
sqrt(9)
sin(pi/2)
log(100)
```

Example output:

```
#0=14
#1=3
#2=1
#3=2
```

Interactive Prompt

The ATC prompt supports command-line editing while typing:

- press Tab to complete documented commands, mathematical functions and user functions;
- press Tab again to cycle through alternative matches when more than one completion is possible;
- use the Up and Down arrows to reuse previous expressions;
- use Left, Right, Home, End, Delete and Backspace for normal line editing.

Autocomplete inserts the closest available match first and can be used more than once in the same expression.

Angle Mode

ATC supports configurable angle modes through the mode command:

```
mode
```

The menu presents:

```
radian -> 1
degree -> 2
gradian -> 3
```

Precision Mode

ATC can persistently switch between double and Boost mp_float:

```
higherprecision(1)
higherprecision(0)
```

The selected mode is applied after restarting ATC.

Useful high-precision examples:

```
dp50dppi
dp50dpe
dp50dmaxprecpi
```

Variables

ATC supports user variables and reusable expressions. Variable-name behavior in 2.1.7 is broader than earlier parser restrictions, while exact reserved names remain protected.

Example:

```
data=5
data+1
```

Matrices

ATC supports matrix commands and matrix-related helper functions.

Example:

```
create matrix(foo\2\2\3)
min(foo)
max(foo)
avg(foo)
```

Polynomial and Equation Commands

Examples:

```
simplify polynomial((x-2)*(x-3))
roots to polynomial(2\3)
solve equation(x^2-5*x+6)
solve quadratic equation(1\_5\6)
solve equations system(1\1\5;1\_1\1)
```

Supported textual polynomial behavior is documented in docs/ATC_2.1.7_DOCUMENTATION.md.

Solver

The solver (. . .) command supports documented numerical solving paths.

Examples:

```
solver(sin(x)-0.5)
solver(x^2-12x-9)
```

Solver behavior depends on expression type, configured angle mode, and the available solver path.

Recipes / Usage Profiles

These recipes give short, practical entry points into ATC without requiring the full user guide first. Outputs are shown only where they are stable across the validated 2.1.7 regression suite.

Engineering

Objective: solve a polynomial equation.

```
solve equation(x^2-5*x+6)
```

Expected stable output:

```
x1=3
x2=2
```

Use this when a supported polynomial form can be solved through ATC's equation solver paths.

Objective: perform a unit conversion.

```
unit conversions
```

Expected output: example placeholder, because this is an interactive menu.

Use this when working through ATC's documented conversion menus.

Objective: run a physics calculation.

```
physics calculations
```

Expected output: example placeholder, because this is an interactive menu.

Use this for documented physics workflows where ATC prompts for the required values.

Objective: create and reuse a matrix.

```
create matrix(foo\2\2\3)
foo+foo
```

Use this when a matrix should be created once and reused in later matrix expressions.

Education

Objective: evaluate arithmetic with operator precedence.

```
2+3*4
```

Expected stable output:

```
#0=14
```

Use this for quick arithmetic checks and classroom examples.

Objective: evaluate a trigonometric identity.

```
sin(pi/2)
```

Expected stable output in radian mode:

```
#0=1
```

Use this when teaching or checking angle-mode-sensitive trigonometry.

Objective: study geometry through the guided menu.

```
geometry calculations
```

Expected output: example placeholder, because this is an interactive menu.

Use this for documented geometry calculations that require prompted inputs.

DSP

Objective: evaluate a normalized signal helper.

```
sinc(0)
```

Expected stable output:

```
#0=1
```

Use this for simple signal-processing checks.

Objective: run a discrete Fourier transform example.

```
fft(1\0\0\0)
```

Use this for documented DSP workflows involving short numeric sequences.

Objective: combine trigonometry with numeric processing.

```
sin(pi/2)+cos(0)
```

Expected stable output:

```
#0=2
```

Use this for compact trigonometric processing expressions.

Statistics

Objective: calculate a mean from a list.

```
avg(2\4\6)
```

Expected stable output:

```
#0=4
```

Use this for quick list-based average calculations.

Objective: calculate minimum and maximum values.

```
min(3\_1\2)  
max(3\_1\2)
```

Expected stable outputs:

```
#0=-1  
#1=3
```

Use this when inspecting small datasets directly in the command line.

Objective: use the statistics menu.

```
statistics calculations
```

Expected output: example placeholder, because this is an interactive menu.

Use this for guided statistical workflows such as variance and standard deviation where available in the menu.

Finance

Objective: open guided financial calculations.

```
financial calculations
```

Expected output: example placeholder, because this is an interactive menu.

Use this for documented financial workflows such as interest and percentage calculations.

Objective: calculate a simple percentage with arithmetic.

```
100*15/100
```

Expected stable output:

```
#0=15
```

Use this for quick percentage checks without entering an interactive menu.

Verbose Resolution

Verbose resolution can be controlled with:

```
verbose resolution  
verbose resolution(1)  
verbose resolution(0)
```

In 2.1.7, verbose output is intended to explain user-facing calculations while avoiding internal startup/menu noise.

More Help

Inside ATC, try:

```
user guide  
current settings
```

For automated coverage of documented behavior, see:

```
tests\ATC_AUTOMATED_TEST_CASES.md  
tests\ATC_USER_GUIDE_COVERAGE.md
```